

Internship Report – Day 12

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Internship Program: ParkNSecure – Smart Parking Solutions

Day 12 Overview:

Today, I trained an Automatic Number Plate Recognition (ANPR) model. The dataset included vehicle images with annotated license plates. The model learned to detect and extract number plates using object detection techniques. It will now be tested for accuracy and optimized for real-time use.

Note:-As I mentioned yesterday its still taking the much time so I kept the model for training still for more accuracy .

Tools Used Today

GEMINI, SONNET, Chatgpt, vs coder.

Task Given:

TO CREATE THE ANPR MODEL TO DETECT THE LETTERS AND NUMBERS PRESENT IN THE VEHICLES NUMBERPLATES.

Approaches:-

The ANPR system was approached using deep learning-based object detection models like YOLO or SSD. Preprocessing included resizing images and annotating number plate regions. The model was trained to detect and localize plates in diverse lighting and angle conditions. Post-processing involves OCR to extract text from the detected plate regions.

Attachments:-

12/19 [01:26<00:49, 7.11s/it]														
Class Images Instances Box(P				R	mAP50	mAP50-95): 68%			13/19 Class Images Instances Box(P					
AP50	mAP50-95): 74%			14/19	[01:40<00:35,		Class	Images	Instances		R	mAP50	mAP50-95): 79%	
Class Images Instances Box(P				R	mAP50	mAP50-95): 84%			16/19 [01:54< Class Images Instances Box(P					
50	mAP50-95): 89%			17/19	[02:16<		Class	Images	Instances		R	mAP50	mAP50-95): 95%	
ges Instances Box(P				R	mAP50	mAP50-95): 100%			19/19 [02:42< Class Images Instances Box(P					
19/19 [02:42<00:00, 8.56s/it]														
all				579	594	0.54	0.533	0.533	0.278					
licence				579	50	0.359	0.12	0.147	0.0464					
plate				579	544	0.721	0.947	0.918	0.51					
Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size								
3/100	0G	1.627	1.022	1.535										
Class Images Instances Box(P				R	mAP50	mAP50-95): 100%								
Class Images Instances Box(P				R	mAP50	mAP50-95): 100%								
all				579	594	0.722	0.64	0.657	0.34					
licence				579	50	0.573	0.32	0.349	0.127					
plate				579	544	0.872	0.96	0.965	0.553					

train: Scanning D:\PARKnsecure\ANPR_3\Dataset\train\labels.cache... 3081 images, 0 b									
val: Scanning D:\PARKnsecure\ANPR_3\Dataset\valid\labels.cache... 579 images, 0 back									
Image sizes 640 train, 640 val									
Using 0 dataloader workers									
Logging results to model\anpr_model									
Starting training for 100 epochs...									
Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size			
1/100	0G	2.873	2.29	2.587	19	640: 100%			
Class		Images	Instances	Box(P	R	mAP50	mAP50-95): 100%		
all		579	594	0.951	0.408	0.472	0.202		
licence		579	50	1	0	0.0939	0.0301		
plate		579	544	0.902	0.816	0.85	0.374		
Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size			
2/100	0G	1.827	1.175	1.79	23	640: 100%	193/193 [40:10<00:00, 12.49s/it]		
Class		Images	Instances	Box(P	R	mAP50	mAP50-95): 100%		
all		579	594	0.54	0.533	0.533	0.278		
licence		579	50	0.359	0.12	0.147	0.0464		
plate		579	544	0.721	0.947	0.918	0.51		
Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size			
3/100	0G	1.627	1.022	1.535	20	640: 100%	193/193 [49:13<00:00, 15.30s/it]		
Class		Images	Instances	Box(P	R	mAP50	mAP50-95): 100%		
all		579	594	0.564	0.563	0.579	0.31		
licence		579	50	0.433	0.153	0.207	0.0704		
plate		579	544	0.695	0.972	0.951	0.55		
Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size			
4/100	0G	1.579	0.9507	1.461	24	640: 65%	125/193 [6:07:12<9:44:52, 516.07s/it]		

I am training the again to get the more accuracy with the 100 epochs.